JUNIOR DEOGRACIAS MAMBO MOTSOU

EMBEDDED SOFTWARE & HARDWARE ENGINEER

Firmware · Microcontrollers · Digital design · Electronics

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LANGUAGES

English ●●●●●
native (anglophone), TOEIC C1 (2024)

French ●●●●○
fluent, professional use (B2)

TECHNICAL SKILLS

Microcontrollers

STM32 (F446RE, G474RE) · **Infineon TLD4020** (Cortex-M23) · **Melexis MLX81116/MLX81143** · **RP2040** (PIO) · **PIC32MZ** · **PIC24** · **ATmega328P**

Firmware

C (bare-metal, register-level CMSIS, HAL/SDK) · **FreeRTOS** · **Zephyr** · **interrupt and timer design** · **ADC acquisition** · **PWM** · **state machines**

Buses & protocols

LIN · **MeLiBu** · **UART** · **SPI** · **I²C** · **AXI4-Stream** · **Bluetooth serial** · **ISO 14443-A** (MIFARE)

Digital design

VHDL (FSMs, pipelined datapaths, bidirectional IOBUF) · **Vivado** (synthesis, implementation, IP integrator, ILA) · **Vitis HLS** · **XSim** · **Artix-7** · **Zynq-7000**

Electronics

Schematic and PCB design (KiCad, Proteus) · **sensor interfacing** (NTC, I²C/SPI sensors) · **evaluation-board bring-up**

Debug & tools

J-Link · **ST-LINK** · **PICkit on-board debugger** · **GDB** · **serial consoles** · **IAR EWARM** · **MPLAB X + Harmony v3/MCC** · **STM32CubeProgrammer** · **Pico SDK** · **Git** · **CMake** · **Ninja** · **West**

Software

Embedded systems engineer (Polytech Dijon, 2026) working across firmware, electronics and digital hardware. At FORVIA I brought up firmware on new automotive silicon (LIN heater controller, MeLiBu LED nodes) and built the software of a nine-gateway lighting demonstrator. I work from datasheets, isolate faults to the hardware or software layer and validate on the bench; VHDL/FPGA design complements this.

Professional experience

Sep. 2025 – Sep. 2026

FORVIA – Faurecia Interior Systems | Embedded Systems Engineer — apprentice (contrat de professionnalisation)

Surface Activation department, France

Firmware bring-up: LIN-controlled heating-element controller (TCU)

- Developed C firmware on the Infineon TLD4020-3ET (Arm Cortex-M23 with integrated LIN): LIN ON/OFF command handling, heater PWM through the LDRV peripheral, NTC-based temperature regulation, fault protection and a state-machine architecture.
- Diagnosed early bring-up blockers: traced a HardFault in device initialisation to a silicon-revision mismatch on the evaluation board (register absent on older silicon), and a lost debug connection to boot-mode jumper and USB power conditions.
- The NTC divider is supplied from the switched heater output, so the ADC node is a 100 Hz square wave during PWM. Synchronised sampling to the PWM on-time (ISR at mid-pulse, 16-sample averaging), which brought measurement scatter within the regulation requirement.

TLD4020-3STD_EVAL board · **IAR Embedded Workbench** · **Infineon MCU Configuration Wizard** · **on-board J-Link** · **printf over LIN** (MobaXterm, 19 200 baud 8N1)

MeLiBu LED driver port: Melexis MLX81116 to MLX81143

- Ported an existing MeLiBu 1.x driver on RP2040 to the MLX81143: extended-mode sub-address encoding (14 LEDs per node), linear 15-bit PWM encoding confirmed against the datasheet and on the hardware, for a 22×7 panel (154 RGB LEDs).
- Fixed bus-timing faults: break generation with busy-wait while interrupts are masked, critical section narrowed to the break to stop UART RX byte loss, TX-FIFO drain to prevent back-to-back frame collisions; corrected the bus return pin.
- Isolated the panel's LED wiring scramble in the hardware-mapping layer, separate from the input convention; used a PIO-based receiver to confirm slave acknowledgements.

RP2040 · **Raspberry Pi Pico SDK** · **PIO state machines** · **C** · **MeLiBu 1.x**

Interior ambient-lighting demonstrator (software owner)

- Extended the control software of an automotive interior demonstration buck from a few LED displays to nine STM32 UART gateways driving WS2812 LED strips across several lighting zones.
- Wrote a multithreaded Python/PyQt5 application: OpenCV decodes MP4 animations, maps logical LED IDs to frame pixels and distributes each frame to one worker thread per gateway at 921 600 baud, with a barrier to keep all nine gateways in step; custom serial frames (ASCII with delta compression).
- Built an Android tablet app (Kotlin, Jetpack Compose) that controls zones, colours and brightness presets over Bluetooth serial using a JSON command protocol with explicit error codes; added a keyboard fallback mode.
- Adapted the STM32 gateway firmware with minimal changes (ID-offset filtering when two gateways share a UART line).

Python 3 · **PyQt5** · **OpenCV** · **Kotlin/Jetpack Compose** · **STM32** (C) · **WS2812** · **UART** · **Bluetooth SPP** · **Git**

Apr. – Aug. 2025

KMUTT — MDT Lab | Embedded Systems Intern

Bangkok, Thailand

- Image-processing work for a precision-agriculture project: fruit classification by transfer learning on VGG16 (Keras/TensorFlow), 98.4 % accuracy on a 4 599-image test set (Fruits-360, 131 classes), plus an OpenCV contour/colour detection pipeline.
- FPGA ramp-up: designed an 8-bit microcontroller in VHDL (12-state fetch–decode–execute controller, ALU with NZVC flags, memory-mapped ROM/RAM/I/O) following the LaMeres reference architecture; simulated in Vivado 2024.2 / XSim.

Python · **TensorFlow/Keras** · **OpenCV** · **VHDL** · **Vivado 2024.2**

Academic & personal projects

POLYTECH DIJON

Python (PyQt5, OpenCV, threading)
C++/Qt
Kotlin (Android, Jetpack Compose)

CERTIFICATION

- Cisco CCNA 1 (July 2024)

2025 – 2026

Real-time thermometer: bare-metal PIC24, then FreeRTOS on PIC32MZ

Polytech, solo then pair · *github.com/teo1747/pic24-32-rt-thermometer*

- V1 (solo): bare-metal PIC24FJ1024GB610 with RTCC timestamps and SPI EEPROM logging. V2 (pair): FreeRTOS pipeline of acquisition, analysis and display tasks on PIC32MZ2048EFH100, linked by queues and binary semaphores; timestamped messages; °C/°F toggle on a button.

Explorer 16/32 **MPLAB X** **XC16/XC32** **Harmony v3/MCC** **FreeRTOS 10.5** **TC1047A sensor** **on-board debugger**

2025 – 2026

ZBT SRAM controller in VHDL

Polytech SoC course, pair · *github.com/teo1747/vhdl-zbt-sram-controller*

- FSM controller for a Micron 512K × 36 ZBT SRAM: single and 4-word burst read/write on a 36-bit bidirectional bus (Xilinx IOBUF); self-checking testbench against the vendor memory model.

VHDL **Vivado** **Zynq-7000 target** **Micron simulation model**

2025 – 2026

Real-time VGA video pipeline on ZedBoard

Polytech, pair · *github.com/teo1747/zedboard-vga-video-pipeline*

- 640 × 480 chain built in IP integrator: Clocking Wizard, Video Timing Controller, AXI4-Stream video IPs, a VHDL inverter and Vitis HLS overlay IPs. Sync signals verified on hardware with ILA; the final demo showed no image because the HLS IPs had no AXI-Lite master to start them, since reworked as free-running (board retest pending).

Zynq-7000 (xc7z020) **Vivado/Vitis HLS 2022.2** **ILA** **XDC constraints**

Nov. 2024 – Jan. 2025

3×3 image filter on FPGA

Polytech, solo · *github.com/teo1747/vhdl-2d-image-filter*

- Streaming averaging and Gaussian (1-2-1/2-4-2/1-2-1) filters on 8-bit pixels: line buffers on the FIFO Generator IP, pipelined adder tree with explicit bit growth (9, 10, then 11 bits); simulated end-to-end on the Lena image, outputs compared with a software reference.

VHDL **Vivado 2024.1** **Artix-7** (xc7a100t) **XSim** **MATLAB**

PERSONAL

Ongoing. Validation so far is in emulation (QEMU, EmbSIM), not on physical boards.

Oct. 2026 –

EmbLinkRTOS — preemptive RTOS for microcontrollers

personal · *github.com/EmbLinkOs/EmbLinkRTOS*

- Fixed-priority preemptive scheduler (bitmap ready lists, FIFO within a priority); Cortex-M port with PendSV/PSP context switch, lazy FPU save and BASEPRI critical sections; RISC-V (mret, CLINT) and AVR ports.
- Semaphores, mutexes with priority inheritance and priority ceiling, timeouts; conformance tests and differential testing against a Python reference model of the scheduler.
- CMake/Kconfig build with board definitions (NUCLEO-F446RE, Arduino Uno, ESP32-C3); GitHub Actions CI with ASan/UBSan/TSan, QEMU and EmbSIM runs.

Oct. 2026

RFID/PIN access-control terminal on FreeRTOS

personal · *github.com/teo1747/stm32-freertos-access-control*

- Register-level drivers (no vendor HAL) on STM32F446RE: MFRC522 RFID on SPI, SSD1306, DS3231 RTC and 24LC256 EEPROM on I²C, keypad, servo PWM, watchdog; six FreeRTOS tasks with queues, mutexes, event groups and stream buffers; salted, stretched SHA-256 PINs; CRC-checked log.
- Host unit tests with sanitizers and system scenarios in EmbSIM; not yet tested on a physical board.

NUCLEO-F446RE **FreeRTOS 11** **CMSIS** **arm-none-eabi-gcc** **CMake** **Ninja**

Education

2023 – Sep. 2026

Engineering degree (Diplôme d'ingénieur, Master's level) — Embedded Systems | Polytech Dijon

Dijon, France

Graduated. Final-year apprenticeship at FORVIA. Coursework: electronics, PCB design, real-time systems, SoC/FPGA.

2021 – 2023

Two-year preparatory programme — mathematics, physics, electronics | IUC

Cameroon